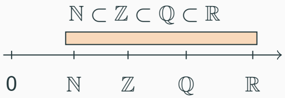
Math Bootcamp

Day 1 — Foundations (Sections 1.1–1.10)

Math Camp 2026

Symbol	English name	What it is
$\mathbb{N}$	natural numbers	Non-negative integers for **counting**
$\mathbb{Z}$	integers	$\ldots, -2, -1, 0, 1, 2, \ldots$
$\mathbb{Q}$	rational numbers	Fractions p/q $(q \neq 0)$
$\mathbb{R}$	real numbers	All points on the number line
$\mathbb{R}^n$	n-dimensional real space	Lists / vectors of length n



1.1 Symbols and vectors — names and definitions

Common symbols

$\in$	"is an element of"	$\forall$	"for all"
$\exists$	"there exists"	$\Rightarrow$	"implies"
		$\Leftrightarrow$	"if and only if" (iff)

Scalar vs. vector

- **Scalar:** one number, e.g. income $w \in \mathbb{R}$
- **Vector:** a list of numbers $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$
- **Dot product:** $\mathbf{p} \cdot \mathbf{x} = \sum_{i=1}^{n} p_i x_i$

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$\rightarrow \mathbb{N}$

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$\to (8.50, 2) \in \mathbb{Q} \times \mathbb{N}$

1.2 Logic — names and definitions

English	Symbol	Meaning
P **only if** Q	$P \Rightarrow Q$	Q is **necessary** for P (without Q, no P)
P **if** Q	$Q \Rightarrow P$	Q is **sufficient** for P (Q alone is enough)
P **if and only if** Q	$P \Leftrightarrow Q$	Q is necessary **and** sufficient
Contrapositive	$\neg Q \Rightarrow \neg P$	same truth as $P \Rightarrow Q$: if Q fails, P must fail

First week in HK: you go to IKEA and spot three plush toys.

Otter HK\$179.9

Hedgehog HK\$79.9

BLÅHAJ HK\$179.9

You have **HK\$500** pocket money ($B = 500$).

Basket $\mathbf{x} \in \mathbb{N}^3$: # otters, # hedgehogs, # sharks. E.g. $(1, 1, 0)$ = one otter, one hedgehog, no shark.
Price vector $\mathbf{p} = (179.9, 79.9, 179.9) \in \mathbb{Q}^3$. Budget set: $\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$.

$\mathbf{p} = (179.9,\ 79.9,\ 179.9)$, $\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$. **True or false?**

$\mathbf{p} = (179.9,\ 79.9,\ 179.9)$, $\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$. **True or false?**

1. $[\in]$ $(0, 0, 3) \in \mathcal{B}$

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1.2 — Plush symbols practice **Together**

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4. $[\exists]$ $\exists\, \mathbf{x} \in \mathcal{B},\ \mathbf{x} = (3, 0, 0)$ **False** $(539.7 > 500)$
5. $[\Rightarrow]$ $(1, 1, 1) \in \mathcal{B}$ **only if** $\mathbf{p} \cdot (1, 1, 1) \leq 500$

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   each affordable $\not\Rightarrow$ jointly affordable

1.3 Sets and operations — names and definitions

Symbol	Name	Meaning
$\in$, $\notin$	element / not in	x belongs (or not) to A
$\subset$, $=$	subset, equality	$A \subset B$; $A = B$ iff mutual $\subset$
$\emptyset$, $\{\,\}$, $\{x \in S : \ldots\}$	sets	empty, roster, set-builder
$A \cup B$, $A \cap B$	union, intersection	in A or B; in both
A^c, $A \setminus B$	complement, difference	not in A; in A not B
$A \times B$	Cartesian product	pairs (a, b), $a \in A$, $b \in B$

In economics: budget set $\mathcal{B} \subset \mathbb{N}^3$; $(8.50, 2) \in \mathbb{Q} \times \mathbb{N}$.

$\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$, $\quad \mathcal{H} = \{\mathbf{x} \in \mathbb{N}^3 : \text{middle entry} \geq 1\}$ (at least one hedgehog).

Describe the set.

$\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}, \quad \mathcal{H} = \{\mathbf{x} \in \mathbb{N}^3 : \text{middle entry} \geq 1\}$ (at least one hedgehog).

Describe the set.

1. $\mathcal{B} \cap \mathcal{H}$

$\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$, $\mathcal{H} = \{\mathbf{x} \in \mathbb{N}^3 : \text{middle entry} \geq 1\}$ (at least one hedgehog).

Describe the set.

1. $\mathcal{B} \cap \mathcal{H}$ affordable baskets with ≥ 1 hedgehog (e.g. $(0, 1, 0)$)

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3. $\mathcal{B} \setminus \mathcal{H}$

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3. $\mathcal{B} \setminus \mathcal{H}$ affordable baskets with **no** hedgehog (e.g. $(1, 0, 0)$)

1.3 — Set operations (IKEA) Together

$\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$, $\mathcal{H} = \{\mathbf{x} \in \mathbb{N}^3 : \text{middle entry} \geq 1\}$ (at least one hedgehog).

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4. $(179.9, 79.9) \in \mathbb{Q} \times \mathbb{Q}$ but $(179.9, 79.9) \notin \mathcal{B}$

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1.3 — Set operations (IKEA) Together

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Describe the set.

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5. $(1, 0, 0) \subset \mathcal{B}$

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4. $(179.9,\ 79.9) \in \mathbb{Q} \times \mathbb{Q}$ but $(179.9,\ 79.9) \notin \mathcal{B}$ True Cartesian product $\neq$ budget set — pairs $\neq$ baskets
5. $(1, 0, 0) \subset \mathcal{B}$ **False** $(1, 0, 0)$ is a basket: use $(1, 0, 0) \in \mathcal{B}$ (True); but $\{(1, 0, 0)\} \subset \mathcal{B}$ is True

1.4 Functions — names and definitions

Term	Notation	Meaning
Function	$f : A \to B$	each $x \in A$ maps to **exactly one** $f(x) \in B$
Domain	A	allowed inputs
Codomain	B	target set — every output must lie in B
Range	$f(A)$	outputs hit: $\{f(x) : x \in A\} \subseteq B$
Injective (one-to-one)		$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$
Surjective (onto)		every $y \in B$ equals some $f(x)$
Operations	$(f \pm g)(x)$, $(cf)(x)$, $(fg)(x)$	$(f \pm g)(x) = f(x) \pm g(x)$; $(cf)(x) = c\,f(x)$; $(fg)(x) = f(x)g(x)$
Composition	$(f \circ g)(x)$	apply g first: $(f \circ g)(x) = f(g(x))$

Define $f : D \to C$ by: $f(x) =$ your girlfriend / boyfriend's age when **your age** is x.

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7. If $f(25) = \{22, 80\}$, is f a function? **No — a correspondence** (a function gives **exactly one** output per input; it's immoral to have a correspondence ☺)

1.5 — Reservation price (example first) Together

BLÅHAJ sticker price: **HK\$179.9**. You buy **only on sale**: when $p < 179.9$.

Prices where you still buy: $P = \{p \in \mathbb{R}_+ : p < 179.9\}$.

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Reservation price (WTP)? $\sup P = 179.9$ least upper bound — the choke price)

Is $179.9 \in P$**?** **No** at sticker price you walk away $\Rightarrow$ no $\max P$)

1.5 Supremum and infimum — names and definitions

Term	Notation	Meaning
Maximum	$\max S$	largest element **in** S
Minimum	$\min S$	smallest element **in** S
Supremum	$\sup S$	**least upper bound** of S
Infimum	$\inf S$	**greatest lower bound** of S

Example (least upper bound): $S = [0, 1)$. Upper bounds include $1, 2, 3, \ldots$; $\sup S = 1$ — the **smallest** upper bound ($1 \notin S$).

Relations: if $\max S$ exists, $\max S = \sup S$; if $\min S$ exists, $\min S = \inf S$.

Always: $\inf S \leq x \leq \sup S$ for all $x \in S$.

1.6 Taylor, logs, and exponentials

Taylor expansion (around a):

$$f(a+h) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}\, h^k = f(a) + f'(a)h + \frac{f''(a)}{2!}h^2 + \frac{f^{(3)}(a)}{3!}h^3 + \cdots$$

Logs

$\ln(ab) = \ln a + \ln b$

$\ln(x^\alpha) = \alpha \ln x$

$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$

Exponentials

$e^{a+b} = e^a e^b$

$e^{\alpha x} = (e^x)^\alpha$

$e^x = 1 + x + \frac{x^2}{2!} + \cdots$

Quick check: $\ln(x_1^\alpha x_2^\beta) = \alpha \ln x_1 + \beta \ln x_2$; if $u = x_1^\alpha x_2^{1-\alpha}$, then $\ln u = \alpha \ln x_1 + (1-\alpha)\ln x_2$.

1.6 Why logs?

- **Taylor:** $f(a+h) \approx f(a) + f'(a)h$.
- **Apply to logs:** $\ln(1+x) \approx x$ for small x.
- **Percentage changes:** $\ln(x+\Delta x) - \ln x = \ln\bigl(1+\frac{\Delta x}{x}\bigr) \approx \frac{\Delta x}{x}$, so $\Delta \ln x$ tracks **% change** in x.
- **Example:** $\frac{\Delta x}{x} = 5\% \Rightarrow \Delta \ln x \approx 0.05$.
- **Log–log regression:** $\ln y = \alpha + \beta \ln x$.
- $1\% \uparrow$ in $x \Rightarrow \beta\%$ change in y.
- **Elasticity:** $\varepsilon_{y,x} = \dfrac{\partial y}{\partial x}\dfrac{x}{y} = \dfrac{\Delta y/y}{\Delta x/x} = \dfrac{\partial \ln y}{\partial \ln x}$; in $\ln y = \alpha + \beta \ln x$, $\varepsilon_{y,x} = \beta$.

1.6 Cobb–Douglas — definition

Utility / production:

$$U(x_1, x_2) = x_1^{\alpha} x_2^{1-\alpha}, \qquad Y(K, L) = AK^{\alpha} L^{1-\alpha}, \quad \alpha \in (0, 1).$$

In logs (linear): $\ln U = \alpha \ln x_1 + (1-\alpha) \ln x_2$; $\ln Y = \ln A + \alpha \ln K + (1-\alpha) \ln L$.

Constant elasticity: $\varepsilon_{U,x_1} = \dfrac{\partial \ln U}{\partial \ln x_1} = \alpha$, $\varepsilon_{U,x_2} = 1 - \alpha$; $\varepsilon_{Y,K} = \alpha$, $\varepsilon_{Y,L} = 1 - \alpha$. Same numbers at every (x_1, x_2) or (K, L).

1.6 Why Cobb–Douglas? — first-order approximation

Any smooth utility $u(x_1, x_2) > 0$: define $f(\ln x_1, \ln x_2) = \ln u(x_1, x_2)$.

First-order Taylor around $(\ln \bar{x}_1, \ln \bar{x}_2)$:

$$\ln u(x_1, x_2) \approx f(\ln \bar{x}_1, \ln \bar{x}_2) + \frac{\partial f}{\partial \ln x_1}\bigg|_{\bar{\mathbf{x}}}(\ln x_1 - \ln \bar{x}_1) + \frac{\partial f}{\partial \ln x_2}\bigg|_{\bar{\mathbf{x}}}(\ln x_2 - \ln \bar{x}_2).$$

Let $\alpha_i = \dfrac{\partial \ln u}{\partial \ln x_i}\bigg|_{\bar{\mathbf{x}}}$ and absorb constants into c:

$$\ln u(x_1, x_2) \approx c + \alpha_1 \ln x_1 + \alpha_2 \ln x_2 \quad \Leftrightarrow \quad u(x_1, x_2) \approx e^c\, x_1^{\alpha_1} x_2^{\alpha_2}.$$

Near $\bar{\mathbf{x}}$ only; Cobb–Douglas is exact when $\ln u$ is globally affine in logs.

1.6 Cobb–Douglas — homogeneity and CRS

Homogeneous of degree k: $f(t\mathbf{x}) = t^k f(\mathbf{x})$ for $t > 0$.

Utility $U = x_1^{\alpha} x_2^{1-\alpha}$: $U(tx_1, tx_2) = t^{\alpha} t^{1-\alpha} U(\mathbf{x}) = t\, U(\mathbf{x}) \Rightarrow$ degree 1.

Production $Y = AK^{\alpha} L^{1-\alpha}$: $Y(tK, tL) = t^{\alpha} t^{1-\alpha} Y(K, L) = t\, Y(K, L)$.

CRS (constant returns to scale): double all inputs $\Rightarrow$ double output (homogeneous of degree 1).

1.6 Cobb–Douglas — constant expenditure shares

Max $U = x_1^{\alpha} x_2^{1-\alpha}$ s.t. $p_1 x_1 + p_2 x_2 = w$.

FOC (marginal utility per dollar equal): $\dfrac{\partial U/\partial x_1}{\partial U/\partial x_2} = \dfrac{p_1}{p_2} \Rightarrow \dfrac{\alpha/x_1}{(1-\alpha)/x_2} = \dfrac{p_1}{p_2} \Rightarrow \dfrac{p_1 x_1}{p_2 x_2} = \dfrac{\alpha}{1-\alpha}$.

Budget binds: $p_1 x_1 + p_2 x_2 = w$. Substitute $p_1 x_1 = \dfrac{\alpha}{1-\alpha} p_2 x_2$:

$$w = p_2 x_2 \left(\frac{\alpha}{1-\alpha} + 1 \right) = \frac{p_2 x_2}{1-\alpha} \Rightarrow p_2 x_2 = (1-\alpha) w, \quad p_1 x_1 = \alpha w.$$

Expenditure shares: $\dfrac{p_1 x_1}{w} = \alpha$, $\dfrac{p_2 x_2}{w} = 1 - \alpha$ (constant).

1.6 Why Cobb–Douglas? — summary

Why economists use Cobb–Douglas:

- **Logs = % changes:** elasticities are $\partial \ln y / \partial \ln x$.
- **Local approximation:** any smooth $u > 0$ satisfies $\ln u \approx c + \alpha_1 \ln x_1 + \alpha_2 \ln x_2$ near $\bar{\mathbf{x}}$.
- **Constant elasticity / shares:** $\varepsilon_{u,x_i} = \alpha_i$; CD demands give fixed budget shares.
- **Homogeneity (degree 1) / CRS:** $f(t\mathbf{x}) = tf(\mathbf{x})$; double inputs $\Rightarrow$ double output.
- **Easy algebra:** linear in logs, powers in levels; closed-form demands and FOCs.
- **Global use = modeling choice:** tractability and homogeneity, not a theorem for all preferences.

1.7 Single-variable derivative — formal definition

Scope: one independent variable. Partial derivatives ($\partial f/\partial x_i$) come later.

Definition. For $f : \mathbb{R} \to \mathbb{R}$ and $a \in \mathbb{R}$, the **derivative at** a is

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h},$$

provided the limit exists. Same number as $\left.\dfrac{df}{dx}\right|_{x=a}$.

Meaning: instantaneous rate of change; slope of the tangent at $(a, f(a))$.

In economics: $MC = C'(q)$, $MU = U'(x)$, $MP_K = Y'(K)$.

Single-variable calculus: all rules below are for $f(x)$, $g(x)$, etc. Built from $f'(a) = \lim_{h\to 0} \dfrac{f(a+h) - f(a)}{h}$ (derivations in appendix):

$(c)' = 0$	constant
$(f \pm g)' = f' \pm g'$	sum / difference
$(x^n)' = nx^{n-1}$	power rule
$(fg)' = f'g + fg'$	product rule
$\left(\frac{f}{g}\right)' = \dfrac{f'g - fg'}{g^2}$	quotient rule
$(e^x)' = e^x$, $(\ln x)' = \frac{1}{x}$	exp / log
$(a^x)' = a^x \ln a$	general exponential ($a > 0$)
$(f^{-1})'(y) = \dfrac{1}{f'(f^{-1}(y))}$	inverse function

Chain rule (composition): if $y = f(u)$, $u = g(x)$, then $\dfrac{dy}{dx} = \dfrac{dy}{du} \cdot \dfrac{du}{dx}$.

1.7 Single-variable chain rule

Chain rule: if $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = (f' \circ g)(x)\, g'(x).$$

Example: $y = (3x+1)^5$. Let $u = 3x+1$: $\dfrac{dy}{dx} = 5u^4 \cdot 3 = 15(3x+1)^4$.

Differentiate (one variable at a time).

1. Cubic cost: $C(q) = q^3 - 6q^2 + 15q + 100$. $C'(q) = 3q^2 - 12q + 15$ MC

2. Monopolist: $p(q) = 120 - 2q$, $C(q) = q^2 + 10q + 50$. Profit $\pi(q) = q(120 - 2q) - C(q) = 110q - 3q^2 - 50$.
$\pi'(q) = 110 - 6q$ FOC: $q^* = 110/6 \approx 18.3$

3. Cobb–Douglas production: $Y(K) = 3K^{0.6}$. $Y'(K) = 1.8\,K^{-0.4}$ MP_K

4. Fixed cost + composite cost: $TC(q) = 100 + 20(2q + 1)^3$. $TC'(q) = 120(2q + 1)^2$ chain rule

1.7 Euler's theorem

Setup (from 1.6). $f : \mathbb{R}^n \to \mathbb{R}$ is C^1, homogeneous of degree k: $f(t\mathbf{x}) = t^k f(\mathbf{x})$ for $t > 0$.

Euler's theorem. $\displaystyle\sum_{i=1}^{n} x_i \frac{\partial f}{\partial x_i}(\mathbf{x}) = k\, f(\mathbf{x})$.

Proof. Fix $\mathbf{x}$, $\varphi(t) = f(t\mathbf{x}) = t^k f(\mathbf{x})$.

RHS: $\varphi'(t) = k\, t^{k-1} f(\mathbf{x})$. **LHS (chain rule):** $\varphi'(t) = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(t\mathbf{x})\, x_i$.

At $t = 1$: $\sum_i x_i f_{x_i}(\mathbf{x}) = kf(\mathbf{x})$.

Multivariate chain rule; partials defined formally in 1.8.

1.7 Euler's theorem — the adding-up problem

Economic question. In a competitive economy, if labor and capital are each paid its **marginal product**, do total factor payments exactly equal total output? Or is there surplus profit or a deficit?

This is the **adding-up problem** (Clark, Wicksteed). If payments do not exhaust output, marginal productivity theory of distribution looks inconsistent.

Mathematical condition. Production $Q = f(K, L)$, CRS (degree 1): $f(tK, tL) = t\, f(K, L)$ for $t > 0$. Euler's theorem ($k = 1$):

$$Q = \frac{\partial Q}{\partial K}\, K + \frac{\partial Q}{\partial L}\, L = MP_K \cdot K + MP_L \cdot L.$$

Algebra only so far — no claim yet about wages or rents.

1.7 Euler's theorem — competitive factor markets

Total factor payments: $rK + wL$.

Perfect competition in factor markets $\Rightarrow$ each factor is paid its **value marginal product**:

$$r = P\frac{\partial Q}{\partial K}, \qquad w = P\frac{\partial Q}{\partial L}.$$

Euler (CRS, $k = 1$). $Q = \dfrac{\partial Q}{\partial K}K + \dfrac{\partial Q}{\partial L}L$. Multiply by P:

$$PQ = rK + wL.$$

Revenue equals factor payments — output **exactly exhausted**; zero profit.

If returns to scale $\neq 1$, see IRS/DRS next.

1.7 Euler's theorem — IRS and DRS

CRS ($k = 1$). $PQ = rK + wL$ — payments exhaust revenue; zero profit.

IRS ($k > 1$). Paying each factor its marginal product would cost **more** than revenue — positive profit wedge (or inconsistency with perfect competition).

DRS ($k < 1$). MP payments **fall short** of revenue — output not fully exhausted; surplus remains.

Intuition: under IRS, marginal contributions add up to more than current output; under DRS, to less.

1.7 Euler's theorem — why IRS implies losses

Better technology, IRS. Production has **synergy** among inputs: at a given scale, doubling all inputs more than doubles output — the extra is synergy.

Marginal product traps synergy. $MP_K = \partial Q/\partial K$ holds L (and other inputs) **fixed** at current levels. Those inputs are already in place, so an extra unit of K gets credit for synergy between the new capital and existing labor.

Repeat for every machine and every worker: the **same synergy** is credited to each input. Summing $MP_K \cdot K + MP_L \cdot L$ therefore **counts synergy several times**.

Euler. Under IRS ($k > 1$), $MP_K \cdot K + MP_L \cdot L = k\,Q > Q$. If each factor is paid its MP, total payments exceed revenue — the firm would run **losses** under perfect competition.

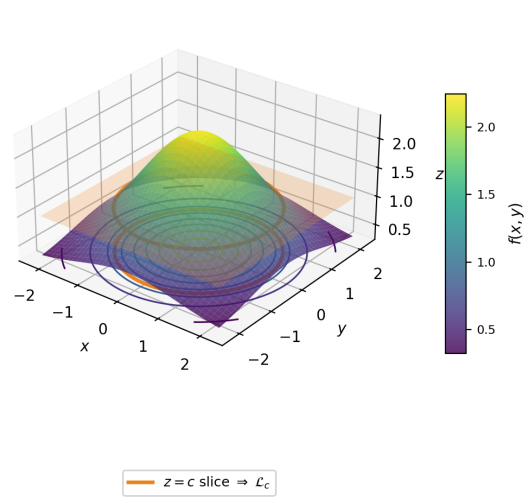
Multivariate function: $z = f(x,y)$, e.g. utility $U(x_1, x_2)$.

Level curve at height c:

$$\mathcal{L}_c = \{(x,y) \in \mathbb{R}^2 : f(x,y) = c\}.$$

On a hill/mountain graph, slice the surface by the horizontal plane $z = c$; the slice projects to a level curve in the (x,y)-plane.

Contour map: top-down picture of several $\mathcal{L}_c$'s. Along a level curve, f is **constant**.



1.8 Multivariate calculus — roadmap

Question: how does $f(x, y)$ change when inputs move?

- Hold one variable fixed $\Rightarrow$ **partial derivatives**
- Small joint change (dx, dy) $\Rightarrow$ **total differential**
- Move in a chosen direction $\mathbf{u}$ $\Rightarrow$ **directional derivative** (rate version of df)
- Differentiate in two orders $\Rightarrow$ **cross-partials** (Schwarz)

1.8 Partial derivatives

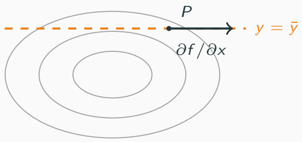
Definition. For $f(x, y)$,

$$\frac{\partial f}{\partial x}(a, b) = \lim_{h \to 0} \frac{f(a+h, b) - f(a, b)}{h},$$

holding $y = b$ fixed (and analogously $\partial f / \partial y$).

On a contour map: fix $y = \bar{y}$ (horizontal line). The slope along that line is $\partial f / \partial x$.

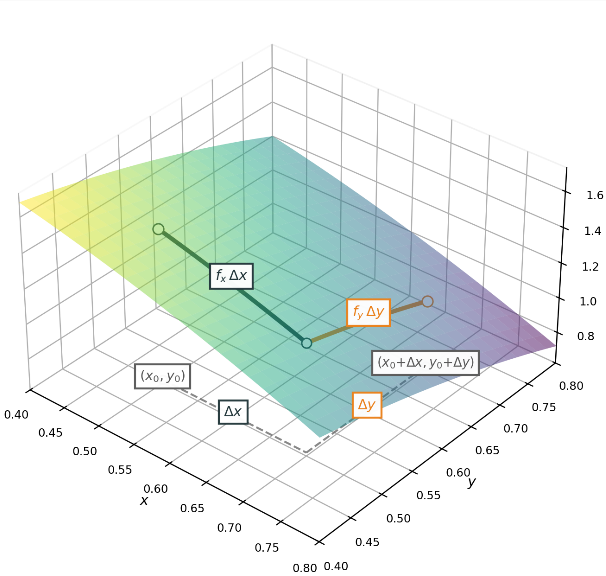
Econ: $MP_K = \partial Y / \partial K$ with L fixed.



1.8 Total differential

Goal: approximate how f changes when (x, y) moves a little from (x_0, y_0) by $(\Delta x, \Delta y)$.

1. **Exact change.** $\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$.
2. **Add/subtract** at $(x_0 + \Delta x, y_0)$:
$$\Delta z = \left[f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0 + \Delta x, y_0)\right] + \left[f(x_0 + \Delta x, y_0) - f(x_0, y_0)\right].$$
3. **Mean value theorem** $(\theta_1, \theta_2 \in (0, 1))$:
$$\begin{aligned} f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0 + \Delta x, y_0) &= f_y(x_0 + \Delta x, y_0 + \theta_1 \Delta y)\,\Delta y, \\ f(x_0 + \Delta x, y_0) - f(x_0, y_0) &= f_x(x_0 + \theta_2 \Delta x, y_0)\,\Delta x. \end{aligned}$$
4. **Small steps** $\Delta x, \Delta y \to 0$: $\Delta z \approx f_x\,\Delta x + f_y\,\Delta y =: dz, \quad \nabla f = (f_x, f_y)$.



$f_x\,\Delta x$
$f_y\,\Delta y$
(x_0, y_0)
Δx
Δy
$(x_0+\Delta x, y_0+\Delta y)$
x
y
0.40
0.45
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.8
1.0
1.2
1.4
1.6

1.8 Using the total differential

When: approximate how output changes when **several inputs** move a little (comparative statics, linearization).

Example. Cobb–Douglas production $Y = 3K^{\alpha}L^{1-\alpha}$ with $\alpha = 0.5$, at $(K_0, L_0) = (100, 64)$, $dK = 2$, $dL = 1$:

$$dY = \underbrace{1.5K^{-0.5}L^{0.5}}_{MP_K}\, dK + \underbrace{1.5K^{0.5}L^{-0.5}}_{MP_L}\, dL.$$

$$dY = \frac{1.5}{10} \cdot 8 \cdot 2 + \frac{1.5 \cdot 10}{8} \cdot 1 = 2.4 + 1.875 \approx 4.28.$$

Exact: $Y(100, 64) = 240$, $Y(102, 65) = 3\sqrt{6630} \approx 244.27$, so $\Delta Y \approx 4.27$ (close to dY when dK, dL are small).

1.8 Gradient and level curves

At P on $\mathcal{L}_c$: $\nabla f(P) \perp \mathcal{L}_c$ and points **uphill**.

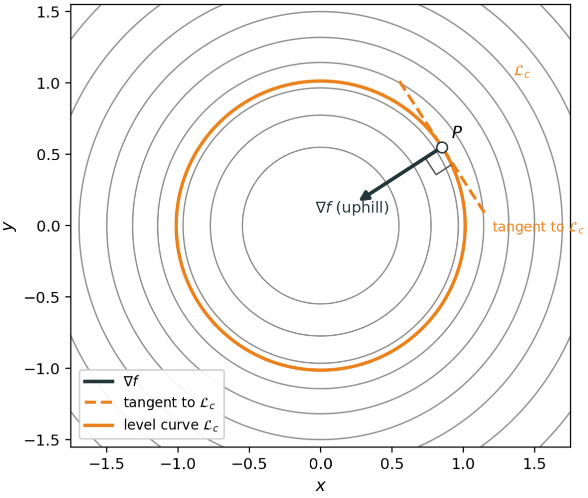
1. Perpendicular. Path on the curve: $\mathbf{r}(t) = (x(t), y(t))$; tangent $\mathbf{r}'(t) = (x'(t), y'(t))$. On $\mathcal{L}_c$, $f(\mathbf{r}(t)) = c$, so

$$0 = \frac{d}{dt} f(\mathbf{r}(t)) = \nabla f \cdot \mathbf{r}'(t) \Rightarrow \nabla f \perp \mathbf{r}'(t).$$

2. Uphill. For a small move $\mathbf{h} = (dx, dy)$,

$$df \approx f_x\, dx + f_y\, dy = \nabla f \cdot \mathbf{h}$$
$$= \sqrt{dx^2 + dy^2}\, \sqrt{f_x^2 + f_y^2}\, \cos\theta,$$

with θ the angle between $\mathbf{h}$ and ∇f. ∇f is uphill: $\mathbf{h} \parallel \nabla f \Rightarrow \cos\theta = 1$, so df is largest and positive.



1.8 Directional derivative

Definition. For unit direction $\mathbf{u} = (u_1, u_2)$, $\|\mathbf{u}\| = 1$, the **directional derivative** at (a, b) is

$$D_{\mathbf{u}} f(a, b) := \lim_{t \to 0} \frac{f(a + tu_1,\ b + tu_2) - f(a, b)}{t}$$

(rate of change of f per unit step along $\mathbf{u}$).

Derive. Step $(dx, dy) = t\mathbf{u}$. By the total differential, for small t,

$$f(a + tu_1, b + tu_2) - f(a, b) \approx f_x(tu_1) + f_y(tu_2) = t\,\nabla f \cdot \mathbf{u}.$$

Divide by t and let $t \to 0$:

$$D_{\mathbf{u}} f(a, b) = \nabla f \cdot \mathbf{u} = \|\nabla f\| \cos\theta.$$

1.8 Example: targeted input change — question

A firm's output with a targeted input change.

Cobb–Douglas production (capital K, labor L):

$$Q(K, L) = K^{\alpha} L^{1-\alpha}, \qquad \alpha = 0.5.$$

Current inputs: $(K_0, L_0) = (100, 100)$.

A subsidy is paid only if the firm increases **both** inputs in ratio 2:1 (for every 1 unit of labor, add 2 units of capital).

This defines direction $\mathbf{v} = (2, 1)$ in (K, L)-space.

Question: How fast does output Q change per unit step in that direction?

1.8 Example: targeted input change — solution

1. Gradient at $(100, 100)$. $Q_K = 0.5K^{-0.5}L^{0.5}$, $Q_L = 0.5K^{0.5}L^{-0.5}$. Since $100^{0.5} = 10$ and $100^{-0.5} = 0.1$,

$$Q_K = 0.5(0.1)(10) = 0.5, \qquad Q_L = 0.5(10)(0.1) = 0.5.$$

So $\nabla Q = (0.5,\ 0.5)$.

2. Unit direction. $\|\mathbf{v}\| = \sqrt{5}$,

$$\mathbf{u} = \left(\tfrac{2}{\sqrt{5}},\ \tfrac{1}{\sqrt{5}}\right).$$

3. Directional derivative.

$$D_{\mathbf{u}}Q = \nabla Q \cdot \mathbf{u} = 0.5\left(\tfrac{2}{\sqrt{5}} + \tfrac{1}{\sqrt{5}}\right) = \frac{1.5}{\sqrt{5}} \approx 0.67.$$

Interpretation: from $(100, 100)$, moving in direction $\mathbf{u}$ (ratio 2:1), output rises at initial rate ≈ 0.67 units of Q per unit distance in input space.

1.8 Cross-partial derivatives and Schwarz

Cross-partials. For $f(x, y)$,

$$f_{xy} = \frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right), \qquad f_{yx} = \frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right).$$

Schwarz's theorem (stated). If f_{xy} and f_{yx} exist and are continuous near (a, b), then $f_{xy}(a, b) = f_{yx}(a, b)$.

Example. $Y = K^{\alpha} L^{1-\alpha}$:

$$\frac{\partial MP_K}{\partial L} = Y_{KL} = \alpha(1-\alpha)K^{\alpha-1}L^{-\alpha} = \frac{\partial MP_L}{\partial K} = Y_{LK}.$$

Econ: Y_{KL} tells how MP_K changes when L rises (and vice versa). Sign of Y_{KL}: complements $(+)$ vs. substitutes $(-)$.

1.9 Limits — why this section

Berkeley's critique (1734). Early calculus treated h as "infinitely small": nonzero when dividing, then set to 0. Berkeley called these **"ghosts of departed quantities"** — logically suspicious.

Example: $f(x) = x^2$ at $a = 3$.

$$\frac{f(3+h) - f(3)}{h} = \frac{(3+h)^2 - 9}{h} = \frac{6h + h^2}{h} = 6 + h \qquad (h \neq 0).$$

Early calculus: "let $h \to 0$" to get $f'(3) = 6$.

Berkeley's objection. The algebra **requires** $h \neq 0$ to divide by h; then we act as if $h = 0$ at the end. That is the ghost: h is treated as both present and absent.

1.9 Limit of a sequence (ε-N)

Let (a_n) be a sequence and $L \in \mathbb{R}$.

Definition. $\lim_{n\to\infty} a_n = L$ means:

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ \forall n \geq N : \quad |a_n - L| < \varepsilon.$$

Reading the quantifiers.

- ε: target accuracy (how close to L you want).
- N: step after which the sequence stays inside that accuracy.
- Order matters: you choose ε first, *then* find N.

If the limit exists, the sequence **converges** to L; otherwise it **diverges**.

1.9 Example: $\lim_{n\to\infty} 1/n = 0$

Claim: $\lim\limits_{n\to\infty} \frac{1}{n} = 0$.

Fix any $\varepsilon > 0$. Need N so that $n \geq N$ implies

$$\left|\frac{1}{n} - 0\right| = \frac{1}{n} < \varepsilon \qquad \Leftrightarrow \qquad n > \frac{1}{\varepsilon}.$$

Choose any integer $N > 1/\varepsilon$ (e.g. $N = \lfloor 1/\varepsilon \rfloor + 1$). Then for all $n \geq N$,

$$\frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

So for every accuracy ε, after a large enough N the sequence stays within ε of 0.

1.9 Example: $\lim_{n\to\infty} \frac{2n}{n+1} = 2$

Claim: $\lim\limits_{n\to\infty} \dfrac{2n}{n+1} = 2$. First simplify the distance to the candidate limit:

$$\left|\frac{2n}{n+1} - 2\right| = \left|\frac{2n - 2(n+1)}{n+1}\right| = \frac{2}{n+1}.$$

Fix $\varepsilon > 0$. Need $\dfrac{2}{n+1} < \varepsilon$, i.e. $n+1 > \dfrac{2}{\varepsilon}$. Choose any integer $N > \dfrac{2}{\varepsilon} - 1$ (e.g. $N = \lfloor 2/\varepsilon \rfloor$). Then for all $n \geq N$,

$$\left|\frac{2n}{n+1} - 2\right| = \frac{2}{n+1} \leq \frac{2}{N+1} < \varepsilon.$$

Pattern: guess L, rewrite $|a_n - L|$ as something that clearly goes to 0, then pick N from that bound.

1.9 Limit of a function (ε-δ)

$\lim_{x\to a} f(x) = L$ means $\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in D$: $0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$.

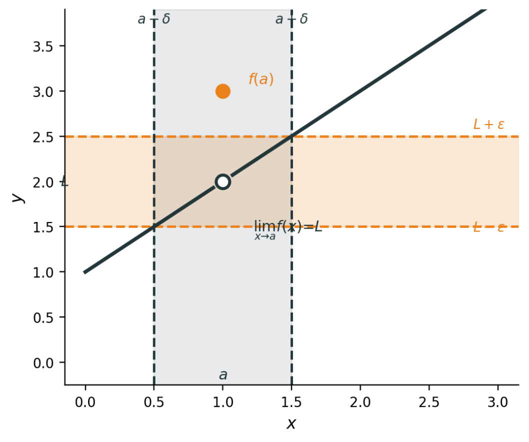
Example. $f(x) = x + 1$ for $x \neq 1$; $f(1) = 3$.

$$\lim_{x\to 1} f(x) = 2, \qquad f(1) = 3 \neq 2.$$

For $\varepsilon = \frac{1}{2}$, take $\delta = \frac{1}{2}$.

If $0 < |x - 1| < \delta$, then

$$|f(x) - 2| = |x - 1| < \varepsilon.$$



On $0 < |x - a| < \delta$, the curve stays in the ε-band around L; the value at $x = a$ is ignored.

1.9 Continuous and differentiable

Continuous at a. f is continuous at a if

$$\lim_{x \to a} f(x) = f(a).$$

So the limit exists **and** equals the function value at a.

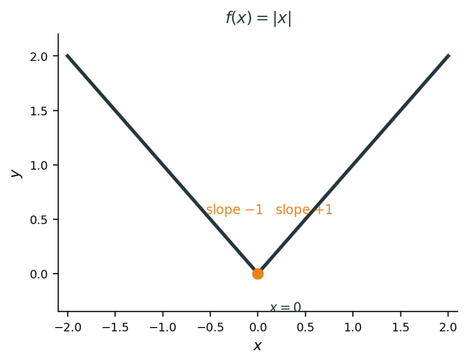
Differentiable at a. f is differentiable at a if

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

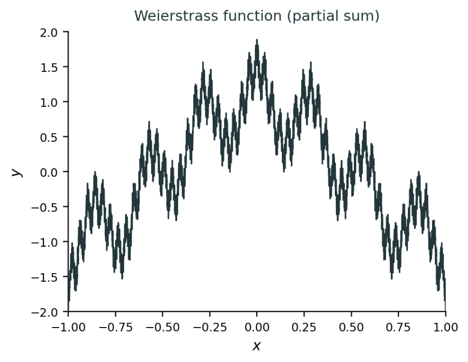
exists (finite).

Relation. Differentiable at $a \Rightarrow$ continuous at a. Converse fails: see examples on the next slide.

1.9 Continuous but not differentiable — for interest



$f(x) = |x|$. Continuous on $\mathbb{R}$; at 0, left slope -1 and right slope $+1$ disagree $\Rightarrow$ not differentiable at 0.



Weierstrass (1872). Some $W : \mathbb{R} \to \mathbb{R}$ is **continuous everywhere** but **differentiable nowhere**. Plot: 10 terms of $\sum_{n=0}^{\infty} a^n \cos(b^n \pi x)$, $a = \frac{1}{2}$, $b = 7$.

1.9 Multivariable: continuous and differentiable

For $f : \mathbb{R}^n \to \mathbb{R}$, write $\mathbf{a} \in \mathbb{R}^n$, $\mathbf{h} \in \mathbb{R}^n$ with $\|\mathbf{h}\| = \sqrt{h_1^2 + \cdots + h_n^2}$.

Continuous at a.

$$\lim_{\mathbf{x} \to \mathbf{a}} f(\mathbf{x}) = f(\mathbf{a}).$$

Must hold as $\mathbf{x} \to \mathbf{a}$ along **every path** (every direction). If two paths give different limits, f is not continuous at $\mathbf{a}$.

Differentiable at a. There is a vector $\nabla f(\mathbf{a}) = (f_{x_1}(\mathbf{a}), \ldots, f_{x_n}(\mathbf{a}))$ such that

$$f(\mathbf{a} + \mathbf{h}) = f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h} + r(\mathbf{h}), \qquad \frac{r(\mathbf{h})}{\|\mathbf{h}\|} \to 0 \text{ as } \mathbf{h} \to \mathbf{0}.$$

Tangent plane (two variables). If $f(x, y)$ is differentiable at (a, b), the graph $z = f(x, y)$ has a tangent plane at $(a, b, f(a, b))$:

$$z - f(a, b) = f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

The linear term $\nabla f(a, b) \cdot \mathbf{h}$ is exactly this plane: the best linear approximation to f near (a, b).

1.9 Differentiable $\Rightarrow$ continuous

Proposition. If $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\mathbf{a}$, then f is continuous at $\mathbf{a}$.

Proof. Write $\mathbf{h} = \mathbf{x} - \mathbf{a}$. By differentiability,

$$f(\mathbf{x}) - f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{h} + r(\mathbf{h}).$$

As $\mathbf{x} \to \mathbf{a}$, we have $\mathbf{h} \to \mathbf{0}$, so $\nabla f(\mathbf{a}) \cdot \mathbf{h} \to 0$ and $r(\mathbf{h}) \to 0$. Hence $f(\mathbf{x}) \to f(\mathbf{a})$. $\square$

1.9 Example: differentiable in two variables

Example. $f(x, y) = x^2 + y^2$ is differentiable at every (a, b).

Check at (a, b). Expand:

$$f(a+h, b+k) - f(a, b) = (a+h)^2 + (b+k)^2 - (a^2 + b^2) = 2ah + 2bk + (h^2 + k^2).$$

Here $\nabla f(a, b) = (2a, 2b)$ and the linear part is $2ah + 2bk$. The remainder is $r(h, k) = h^2 + k^2$, and

$$\frac{|r(h, k)|}{\|(h, k)\|} = \frac{h^2 + k^2}{\sqrt{h^2 + k^2}} = \sqrt{h^2 + k^2} \to 0.$$

So f is differentiable at (a, b) with $df = 2a\, dx + 2b\, dy$.

By the proposition on the previous slide, f is also continuous at (a, b).

1.9 Partials exist $\not\Rightarrow$ differentiable — for interest

Partials at (a, b) alone do **not** imply differentiable, and do **not** imply a tangent plane exists.

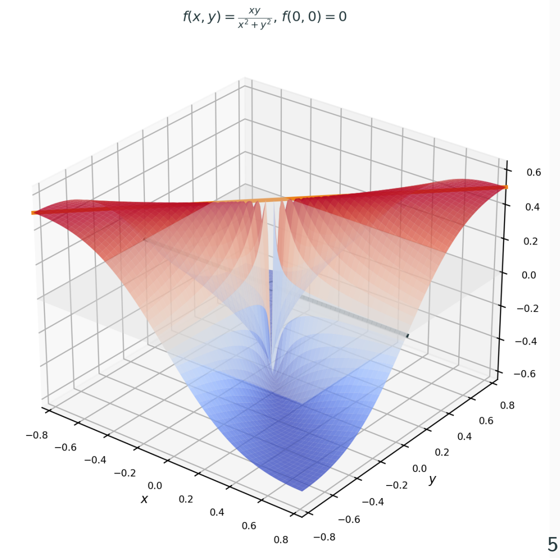
Example. Define

$$f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Partials at $(0,0)$**:** $f(x,0) = 0$, $f(0,y) = 0 \Rightarrow f_x(0,0) = f_y(0,0) = 0$.

But not continuous at $(0,0)$**:** along $y = x$, $f(x,x) = \frac{1}{2}$ for $x \neq 0$; along $y = 0$, $f = 0$. Different paths, different limits.

So no tangent plane: the graph cannot be approximated by one plane from every direction.



Practice.

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Goal. $\forall \varepsilon > 0 \; \exists \delta > 0$: if $0 < |h| < \delta$, then $\left| \frac{f(a+h)-f(a)}{h} - 2a \right| < \varepsilon$.

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Algebra. For $h \neq 0$, $\dfrac{(a+h)^2 - a^2}{h} = \dfrac{2ah + h^2}{h} = 2a + h$.

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2. $f(x) = x + 1$ for $x \neq 1$; $f(1) = 3$. Use ε-δ to prove $\lim\limits_{x \to 1} f(x) = 2$.

Practice.

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Goal. $\forall\, \varepsilon > 0\ \exists\, \delta > 0$: if $0 < |x - 1| < \delta$, then $|f(x) - 2| < \varepsilon$.

Algebra. For $x \neq 1$ near 1, $f(x) = x + 1$, so $|f(x) - 2| = |x - 1|$.

Practice.

1. Let $f(x) = x^2$. Use ε-δ to prove $f'(a) = 2a$.

Goal. $\forall\, \varepsilon > 0\ \exists\, \delta > 0$: if $0 < |h| < \delta$, then $\left|\frac{f(a+h)-f(a)}{h} - 2a\right| < \varepsilon$.

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Choose $\delta = \varepsilon$**.** Then $\left|\frac{f(a+h)-f(a)}{h} - 2a\right| = |h| < \varepsilon$.   so $f'(a) = 2a$

2. $f(x) = x + 1$ for $x \neq 1$; $f(1) = 3$. Use ε-δ to prove $\lim\limits_{x \to 1} f(x) = 2$.

Goal. $\forall\, \varepsilon > 0\ \exists\, \delta > 0$: if $0 < |x - 1| < \delta$, then $|f(x) - 2| < \varepsilon$.

Algebra. For $x \neq 1$ near 1, $f(x) = x + 1$, so $|f(x) - 2| = |x - 1|$.

Choose $\delta = \varepsilon$**.** Then $|f(x) - 2| = |x - 1| < \varepsilon$. $f(1) = 3 \neq 2$: limit exists, but f not continuous at 1

1.10 CES — definition

CES (constant elasticity of substitution) production / utility:

$$f_\rho(x_1, x_2) = \left(\alpha x_1^\rho + (1-\alpha)x_2^\rho\right)^{1/\rho}, \qquad \alpha \in (0,1),\ \rho \neq 0,\ \rho \leq 1.$$

Plan.

1. Show the elasticity of substitution is **constant** (hence the name).
2. Define perfect substitutes and perfect complements.
3. Homogeneity / CRS (practice).
4. Limit cases: $\rho \to 0$ (Cobb–Douglas) and $\rho \to -\infty$ (Leontief).

1.10 Elasticity of substitution — definition

Along an isoquant (f = constant output; or indifference curve if f is utility),

$$\mathrm{MRS} = \frac{f_1}{f_2} = \frac{\partial f/\partial x_1}{\partial f/\partial x_2}.$$

MRS is the rate at which you can trade x_1 for x_2 while keeping f fixed (slope of the isoquant / indifference curve).

Let $r = x_2/x_1$. The **elasticity of substitution** is

$$\sigma = \frac{d \ln r}{d \ln \mathrm{MRS}} = \frac{\%\,\text{change in input ratio}}{\%\,\text{change in MRS}}.$$

1.10 Elasticity of substitution — intuition

Story. Suppose x_2 becomes relatively more valuable at the margin (MRS rises: you need more x_2 to replace one unit of x_1). How much does the firm / consumer switch toward using relatively more x_1 (i.e. lower $r = x_2/x_1$)?

- σ **large:** easy to swap. A small change in relative scarcity / prices moves the mix a lot. Extreme: perfect substitutes ($\sigma = \infty$) — any mix is fine; isoquants are straight lines.
- σ **small:** hard to swap. Mix barely changes even if relative prices move a lot. Extreme: perfect complements ($\sigma = 0$) — must use fixed proportions; L-shaped isoquants.
- Cobb–Douglas sits in between ($\sigma = 1$).

Competitive markets: at an optimum, MRS $=$ relative prices, so σ also says how strongly input ratios respond to price ratios.

1.10 CES: compute MRS

Let $g = \alpha x_1^\rho + (1-\alpha)x_2^\rho$, so $f_\rho = g^{1/\rho}$. Chain rule:

$$f_1 = \frac{1}{\rho} g^{1/\rho - 1} \cdot \alpha\rho\, x_1^{\rho-1} = \alpha\, x_1^{\rho-1}\, g^{1/\rho-1},$$

and similarly $f_2 = (1-\alpha)\, x_2^{\rho-1}\, g^{1/\rho-1}$.

Cancel the common factor $g^{1/\rho-1} > 0$:

$$\mathrm{MRS} = \frac{f_1}{f_2} = \frac{\alpha}{1-\alpha}\left(\frac{x_1}{x_2}\right)^{\rho-1} = \frac{\alpha}{1-\alpha}\, r^{1-\rho}, \qquad r = \frac{x_2}{x_1}.$$

1.10 CES: σ is constant

From $\mathrm{MRS} = \dfrac{\alpha}{1-\alpha}\, r^{1-\rho}$, take logs:

$$\ln \mathrm{MRS} = \ln \frac{\alpha}{1-\alpha} + (1-\rho)\ln r.$$

Differentiate with respect to $\ln r$:

$$\frac{d\ln \mathrm{MRS}}{d\ln r} = 1-\rho \qquad \Rightarrow \qquad \sigma = \frac{d\ln r}{d\ln \mathrm{MRS}} = \frac{1}{1-\rho}.$$

Key: $\sigma = 1/(1-\rho)$ depends only on the parameter ρ, **not** on (x_1, x_2). That is why the family is called CES.

1.10 Perfect substitutes and perfect complements

Perfect substitutes. Goods/inputs that trade off at a **constant** rate: isoquants (or indifference curves) are **straight lines**. Example: $f = \alpha x_1 + (1-\alpha)x_2$ ($\rho = 1$). MRS is constant; $\sigma = \infty$ (any ratio is fine).

Perfect complements. Goods/inputs used in **fixed proportions**: only the bottleneck matters. Leontief form: $f = \min\{x_1, x_2\}$ (L-shaped isoquants). Extra of one input alone is useless; $\sigma = 0$ (no substitution).

CES sits between these extremes via $\sigma = 1/(1-\rho)$.

Let $f_\rho(x_1, x_2) = \left(\alpha x_1^\rho + (1-\alpha)x_2^\rho\right)^{1/\rho}$.

(a) Show f_ρ is homogeneous of degree 1 (CRS: $f_\rho(t\mathbf{x}) = t\, f_\rho(\mathbf{x})$).

(b) For $\rho = 1$, simplify f_ρ. What technology is this?

(c) Using Euler (Day 1.7): what is $x_1 \partial_1 f_\rho + x_2 \partial_2 f_\rho$?

1.10 — CES homogeneity (solution) Together

(a)

$$f_\rho(tx_1, tx_2) = \big(\alpha(tx_1)^\rho + (1-\alpha)(tx_2)^\rho\big)^{1/\rho} = \big(t^\rho(\alpha x_1^\rho + (1-\alpha)x_2^\rho)\big)^{1/\rho} = t\, f_\rho(x_1, x_2).$$

(b) $f_1 = \alpha x_1 + (1-\alpha)x_2$: perfect substitutes (linear isoquants).

(c) Degree 1 $\Rightarrow x_1\partial_1 f_\rho + x_2\partial_2 f_\rho = f_\rho$.

1.10 CES limit $\rho \to 0$ — setup

Fix $x_1, x_2 > 0$. Want $\lim_{\rho \to 0} f_\rho(x_1, x_2)$.

Work with logs (continuous, so $\lim \ln f = \ln \lim f$ when the limit is positive):

$$\ln f_\rho = \frac{1}{\rho} \ln\left(\alpha x_1^\rho + (1-\alpha) x_2^\rho\right) = \frac{\ln g(\rho)}{\rho}, \qquad g(\rho) = \alpha x_1^\rho + (1-\alpha) x_2^\rho.$$

As $\rho \to 0$: $g(0) = \alpha + (1-\alpha) = 1$, so $\ln g(0) = 0$. Numerator $\to 0$, denominator $\to 0$: $0/0$ **form**.

1.10 CES limit $\rho \to 0$ — L'Hôpital

Apply L'Hôpital (differentiate in ρ; write $x_i^\rho = e^{\rho \ln x_i}$):

$$g'(\rho) = \alpha x_1^\rho \ln x_1 + (1-\alpha) x_2^\rho \ln x_2, \qquad \frac{d}{d\rho} \ln g(\rho) = \frac{g'(\rho)}{g(\rho)}.$$

So

$$\lim_{\rho \to 0} \ln f_\rho = \lim_{\rho \to 0} \frac{\ln g(\rho)}{\rho} = \lim_{\rho \to 0} \frac{g'(\rho)}{g(\rho)} = \frac{\alpha \ln x_1 + (1-\alpha) \ln x_2}{1}.$$

Therefore

$$\lim_{\rho \to 0} f_\rho = \exp\big(\alpha \ln x_1 + (1-\alpha) \ln x_2\big) = x_1^\alpha x_2^{1-\alpha} \quad \text{(Cobb–Douglas)}.$$

1.10 CES limit $\rho \to -\infty$ — setup

Fix $x_1, x_2 > 0$. Let $m = \min\{x_1, x_2\}$. Write

$$f_\rho = \left(\alpha x_1^\rho + (1-\alpha)x_2^\rho\right)^{1/\rho} = m\left(\alpha\left(\frac{x_1}{m}\right)^\rho + (1-\alpha)\left(\frac{x_2}{m}\right)^\rho\right)^{1/\rho}.$$

One of x_1/m, x_2/m equals 1; the other is ≥ 1.

Key fact. If $z > 1$ and $\rho \to -\infty$, then $z^\rho = 1/z^{-\rho} \to 0$ (large negative power of a number bigger than 1).

1.10 CES limit $\rho \to -\infty$ — proof

Case $x_1 < x_2$: then $m = x_1$, $x_1/m = 1$, $x_2/m > 1$.

$$\alpha\left(\frac{x_1}{m}\right)^{\rho} + (1-\alpha)\left(\frac{x_2}{m}\right)^{\rho} \xrightarrow[\rho\to-\infty]{} \alpha \cdot 1 + (1-\alpha)\cdot 0 = \alpha.$$

For any $c > 0$, $c^{1/\rho} \to c^0 = 1$ as $\rho \to -\infty$. Hence $f_\rho \to m \cdot 1 = x_1 = \min\{x_1, x_2\}$.

Case $x_2 < x_1$: same argument $\Rightarrow f_\rho \to x_2 = \min\{x_1, x_2\}$.

Case $x_1 = x_2$: inside $\to 1$, so $f_\rho \to x_1 = \min\{x_1, x_2\}$.

Conclusion. $\lim\limits_{\rho\to-\infty} f_\rho(x_1, x_2) = \min\{x_1, x_2\}$ (Leontief / perfect complements).

1.10 CES nesting — summary

Limit / value	Limit function	Interpretation
$\rho = 1$ ($\sigma = \infty$)	$\alpha x_1 + (1-\alpha)x_2$	perfect substitutes
$\rho \to 0$ ($\sigma = 1$)	$x_1^{\alpha} x_2^{1-\alpha}$	Cobb–Douglas
$\rho \to -\infty$ ($\sigma = 0$)	$\min\{x_1, x_2\}$	perfect complements

Same CES family; different ρ (equivalently σ) recover the classic special cases via limits.

A.1 Deriving rules (I): constants, sums, powers

Write $D_a f = f'(a) = \lim_{h \to 0} \dfrac{f(a+h) - f(a)}{h}$.

Constant: $f(x) = c \Rightarrow D_a f = \lim_{h \to 0} \dfrac{c - c}{h} = 0$.

Sum: $D_a(f + g) = D_a f + D_a g$ (split the numerator; limits add).

Power rule (integer $n \geq 1$): for $f(x) = x^n$,

$$\frac{(a+h)^n - a^n}{h} = \frac{a^n + na^{n-1}h + \cdots + h^n - a^n}{h} = na^{n-1} + \text{terms with extra factors of } h \xrightarrow[h \to 0]{} na^{n-1}.$$

So $(x^n)' = nx^{n-1}$.

A.2 Deriving rules (II): product and quotient

Product rule. For $h(x) = f(x)g(x)$,

$$\frac{h(a+h) - h(a)}{h} = f(a+h)\frac{g(a+h) - g(a)}{h} + g(a)\frac{f(a+h) - f(a)}{h} \xrightarrow[h \to 0]{} f(a)g'(a) + f'(a)g(a).$$

So $(fg)' = f'g + fg'$.

Quotient rule. Write $\dfrac{f}{g} = f \cdot g^{-1}$. For $y = 1/g$, $\ln y = -\ln g$, so $\dfrac{y'}{y} = -\dfrac{g'}{g} \Rightarrow y' = -\dfrac{g'}{g^2}$.
Then product rule gives $\left(\dfrac{f}{g}\right)' = \dfrac{f'g - fg'}{g^2}$.

A.3 Key facts for exp and log

Fact 1: $\lim_{h \to 0} \dfrac{e^h - 1}{h} = 1.$

Why? Start with $e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n$ and $e^x = \lim_{n \to \infty} \left(1 + \frac{x}{n}\right)^n$. Binomial expansion:

$$\left(1 + \tfrac{h}{n}\right)^n = 1 + h + \frac{n(n-1)}{2n^2} h^2 + \cdots \xrightarrow[n \to \infty]{} e^h.$$

So $e^h = 1 + h + O(h^2)$, hence $\dfrac{e^h - 1}{h} = 1 + O(h) \to 1.$

Fact 2: $\lim_{u \to 0} \dfrac{\ln(1+u)}{u} = 1.$

Why? For small u, Fact 1 gives $e^u \approx 1 + u$. If $v = \ln(1+u)$, then $e^v = 1 + u$, so $v \approx u$ and $\dfrac{\ln(1+u)}{u} \approx 1$. Equivalently, Taylor: $\ln(1+u) = u - \frac{u^2}{2} + \cdots$.

A.4 Deriving the exponential: $(e^x)' = e^x$

Use Fact 1 from the previous slide: $\lim_{h \to 0} \frac{e^h - 1}{h} = 1$.

For $f(x) = e^x$,

$$f'(a) = \lim_{h \to 0} \frac{e^{a+h} - e^a}{h} = e^a \lim_{h \to 0} \frac{e^h - 1}{h} = e^a.$$

So $(e^x)' = e^x$.

Reading: a small step h multiplies e^x by about $1 + h$.

A.5 Deriving the logarithm: $(\ln x)' = \frac{1}{x}$

Method 1 (inverse function). If $y = \ln x$, then $e^y = x$. Differentiate: $e^y\, y' = 1$, so $y' = \dfrac{1}{e^y} = \dfrac{1}{x}$.

Method 2 (limit definition). For $a > 0$, set $u = h/a$:

$$\frac{\ln(a+h) - \ln a}{h} = \frac{1}{h} \ln\!\left(1 + \frac{h}{a}\right) = \frac{1}{a} \cdot \frac{\ln(1+u)}{u} \xrightarrow[u \to 0]{} \frac{1}{a},$$

by Fact 2. So $(\ln x)' = \frac{1}{x}$.